TUHIN MOHANTA

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PROFESSIONAL SUMMARY

Computer Science and Engineering student who has built and integrated complete web applications across **React.js, Spring Boot, REST APIs, MySQL, PHP and FastAPI**, including a flight booking platform with end-to-end booking and payment workflows and a Generative AI library system using **RAG with FAISS vector search and a locally hosted Llama 3.2 model**. Strong in frontend-backend integration, relational database design, authentication and responsive UI. Seeking Software Developer, Java Developer or Full Stack Developer internship and entry-level roles.

TECHNICAL SKILLS

Programming Languages: Java, Python, C, SQL, JavaScript
Frontend Development: React.js, HTML5, CSS3, Tailwind CSS, Responsive Web Design
Backend Development: Spring Boot, REST APIs, PHP, FastAPI, Authentication
Databases: MySQL, DBMS, Relational Database Design
AI/ML Technologies: RAG, FAISS, Vector Embeddings, Ollama, Llama 3.2, Generative AI, Prompt Engineering, AI Assistant Integration
Developer Tools: Git, GitHub, VS Code, XAMPP
Core Concepts: Object-Oriented Programming (OOP), Debugging, Problem Solving, Software Development Life Cycle

PROJECTS

SkyHigh Flight Booking and E-Commerce System | React.js, Spring Boot, REST APIs, MySQL

- Engineered a full-stack flight booking platform on a React.js frontend, Spring Boot REST backend and MySQL database, covering airport-to-airport search through to ticket issue.
- Implemented seat selection, passenger details, booking and payment workflows, PNR/e-ticket generation, live flight tracking, an admin module and a responsive UI.
- **Key Edge:** Unlike basic flight-search websites, one platform covers search, booking, payment, e-ticket, live tracking and admin control; the layered React, Spring Boot and MySQL design keeps it modular and easy to extend.

GEN-AI Library Management System | React/Vite, Tailwind CSS, PHP, MySQL, FastAPI, FAISS, Ollama

- Developed a multi-service platform with authentication, dashboards, book management, issue/return and approval workflows, history, search, categories and e-books across PHP, MySQL and FastAPI services.
- Engineered an AI RAG pipeline combining vector embeddings, FAISS Top-K similarity search and semantic plus keyword scoring, with answers generated locally by Ollama (Llama 3.2).
- **Key Edge:** Unlike keyword-only library catalogues, it adds semantic AI search and runs the language model locally, so no external AI API is needed and library data stays on the local system.

Weather Intelligence | Spring Boot, REST APIs, JavaScript, AI Assistant

- Developed a responsive weather application with a Spring Boot REST backend serving location-based and coordinate-based weather data through external weather APIs.
- Integrated an AI-powered personal assistant that understands natural-language questions and turns weather data into clear, personalised guidance.
- **Key Edge:** Unlike plain weather websites that only display numbers, it pairs location and coordinate search with an AI assistant that explains conditions in simple language.

Dragon Typing Battle (Typing Test Game) | React.js, JavaScript

- Built an interactive React.js typing game with 100 levels, timer-driven gameplay, difficulty progression, real-time speed (WPM) and accuracy feedback, and performance tracking.
- **Key Edge:** Unlike single-test typing sites, it offers 100 progressive levels that build speed and accuracy step by step.

EDUCATION

B.Tech in Computer Science & Engineering | Ramkrishna Mahato Government Engineering College | 2024 – 2027 (Pursuing) | CGPA: 7.5
Gap Period (2022 – 2024): Preparation for higher studies
Diploma in Civil Engineering | Raipur Government Polytechnic | 2019 – 2022 | 78.7%
Secondary (WBBSE) | Akui Union High School | 2019 | 74.4%

CERTIFICATIONS AND TRAINING

- **AI Foundation (TCS iON Career Edge)** – Tata Consultancy Services | 04 Aug 2026 – 16 Oct 2026
- **AI DevOps Analyst** – NASSCOM via Skill India Digital Hub | Completed 31 Oct 2026 | 1 hour
- **IT Forensic Analyst** – NASSCOM via Skill India Digital Hub | Completed 15 Oct 2026 | 1 hour
- **Bootcamp on Cryptography & Network Security** – IIT Dhanbad | 7 days

LANGUAGES

Bengali (Advanced), English (Advanced), Hindi (Advanced)